Scalable Flash Sale System

Initiation

~~Infrastructure~~

Dawei

Checklist

- [x] VPC, subnets, ElastiCache setup
- [x] ECR + Docker build/push
- [x] ECS Fargate + ALB deployment
- [x] CloudWatch logging

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Planning

~~Core Go Service~~

Rajkesh

Project Plan Checklist

- [x] Purchase endpoint + health check
- [x] In-memory backend
- [x] Redis backend
- [x] Docker + docker-compose

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Execution

Experiment 1 — Horizontal Scaling

Dawei

Checklist

- [ ] Write Locust burst traffic script
- [ ] Run tests at 1, 2, 4 instances
- [ ] Collect and analyze metrics

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Experiment 2 — In-Memory vs Redis

Dawei

Checklist

- [ ] Deploy in-memory variant on ECS
- [ ] Run correctness comparison
- [ ] Document findings

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Experiment 3 — Resilience Patterns

Rajkesh

Checklist

- [ ] Implement Fail Fast middleware
- [ ] Implement Bulkhead middleware
- [ ] Implement Circuit Breaker middleware
- [ ] Run 5,000 user load tests
- [ ] Tune parameters and analyze

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Experiment 4 (TBD)

Rajkesh

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Monitoring

Testing

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Checklist

- [ ] Run Locust against ALB
- [ ] Verify correctness post-run

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Metrics and Logs

Rajkesh

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Closing

Final Deliverables

Unassigned

Dawei + Rajkesh

Checklist

- [ ] Final Report
- [ ] Updated Code Repository
- [ ] Video
- [ ] Presentation

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